

Reinforcement-Learning-Based Optimization on Energy Efficiency in UAV Networks for IoT

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Abstract—The combination of nonorthogonal multiplex access and unmanned aerial vehicles (UAVs) can improve the energy efficiency (EE) for Internet of Things (IoT). On the condition of interference constraint and minimum achievable rate of the secondary users, we propose an iterative optimization algorithm on EE. First, with a given UAV trajectory, the Dinkelbach method-based fractional programming is adopted to obtain the optimal transmission power factors. By using the previous power allocation scheme, the successive convex optimization algorithm is adopted in the second stage to update the system parameters. Finally, reinforcement-learning-based optimization is introduced to obtain the best UAV trajectory.

Index Terms—Energy efficiency (EE), Internet of Things (IoT), nonorthogonal multiplex access (NOMA), power allocation optimization, unmanned aerial vehicles (UAVs).

I. INTRODUCTION

DUE TO the rapid growth of wireless connections, both the total capacity and the spectrum efficiency of Internet of Things (IoT) are facing more and more challenges, including energy efficiency (EE) and QoS limitations. To tackle the bottleneck of the IoT, various new technologies are proposed and investigated in both academic and industrial areas [1], [2], [3], [4], [5].

Unmanned aerial vehicles (UAVs) can be used in IoT to improve the system capacity [6], [7]. Considering energy

harvesting at the user equipments, by using the weight sum method, Kuang et al. [8] introduced a solution to maximize the spectral efficiency. Focusing on hybrid energy supplied networks, Xu et al. [9] proposed a novel wireless resource allocation scheme to improve the long-term mean EE. Considering the presence of an interference node, the suboptimal power splitting and energy allocation method is proposed in [10] to improve the EE of cooperative relay networks. Furthermore, an EE and spectrum efficiency tradeoff scheme are addressed in [11] for the cooperative cognitive radio network, where mathematical analysis as well as the numerical results are present to show that about 19% gains can be obtained compared with the existing methods. Recently, the Dinkelbach method [12]-based optimization on EE is introduced in [13] to solve the fractional programming, where the classical EE problem is transformed into q parameters iterative problem.

Some enabling technologies have been proposed and analyzed to improve the QoS in UAV networks [14], [15]. Considering both the coverage and the EE in UAV networks, with the help of potential games and Nash equilibrium, Ruan et al. [16] developed a UAV deployment scheme to guarantee the energy-efficient coverage. Under the information causality constraint, Ahmed et al. [17] proposed the optimal energy-efficient UAV trajectory for relaying networks. By optimization on locations of backscattering UAV for future IoT, Yang et al. [18] derived the system average outage probability by using the Golden Section method. Considering the wireless-powered UAV networks, by using the Q -learning method, Fu et al. [19] proposed a learning-based UAV route scheme for data collection.

Meanwhile, the deployment of nonorthogonal multiplex access (NOMA) networks is a promising solution for limited spectrum scenarios [20], [21], [22], [23] in cognitive radio. Considering the fairness of multiple users, energy-efficient power allocation is investigated in [24], where only statistical channel state information is used to obtain suboptimal solution under the rate requirements as well as the power constraints. Song and Zheng [25] formulated the EE optimization as a fractional programming problem and proposed a solution method for the nonconvex problem. Focusing on the hybrid user pairing scheme, Nguyen et al. [26] and Wang and Men [27] proposed a joint optimization algorithm on spectral efficiency and EE.

Furthermore, in the latest research literature, machine learning has been adopted in [28] as an enabling algorithm in

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wireless networks. When the traffic load is high enough, the proposed scheme showed higher EE compared with the classical method. Similarly, without prior knowledge, an online network selection algorithm is proposed in [29], [30], and [31] for complex network environment. With imperfect successive interference cancellation (SIC) decoding receiver, both precoding and SIC decoding are jointly optimized in [32] based on deep neural networks. To maximize the sum data rate as well as the EE, a learning-based algorithm is proposed in [33] to tackle the challenge of massive connectivity. A deep reinforcement-learning (RL)-based optimization solution on power and channels for multicarrier NOMA is proposed in [34].

The optimization method on power resource allocation and UAV flight trajectory for NOMA networks (UAV-NOMA) is investigated in this article with interference constraint and minimum achievable rate constraint. In the considered wireless networks, sharing the spectrum of the primary user by using cognitive radio, the UAV node sends multiple data streams simultaneously to its users. By optimization of the UAV movement and the transmission power as well as the power resource allocation scheme, the optimal average EE can be obtained. To solve the difficult problem, an iterative RL-based algorithm on the resource allocation and the flying trajectory is proposed. In the first stage, with a given UAV trajectory, the Dinkelbach method-based fractional programming is adopted to get the optimal transmission power as well as the power allocation scheme. Afterward, by using the previous power allocation scheme, the successive convex optimization algorithm is used in the second stage to update the system parameters. Finally, RL-based optimization is introduced to obtain the optimal UAV movement directions in terms of mean EE. The contributions of this article are summarized as follows.

- 1) Under the constraints of maximum interference power and minimum achievable rate, we investigate the joint optimization algorithm on UAV flight trajectory and power allocation for UAV-NOMA networks in the presence of multiple secondary users.
- 2) Considering the interference constraint as well as the minimum achievable rate of the secondary users, Dinkelbach fractional programming and successive convex optimization algorithm are adopted to solve the fractional resource allocation problem, and the RL-based update algorithm is introduced to solve the UAV trajectory optimization problem.

II. NETWORK MODEL

The system model of UAV-NOMA is shown in Fig. 1, where a primary user, K secondary users, and a single UAV node share the same wireless spectrum as a cognitive network. It is assumed that all wireless links follow line-of-sight propagation, and all nodes are configured with only one antenna. Because of the spectrum-sharing protocol, the signals transmitted by the UAV node will interfere with the primary user. To guarantee the normal data transmission of the primary users, the UAV node must strictly control its own transmit power.

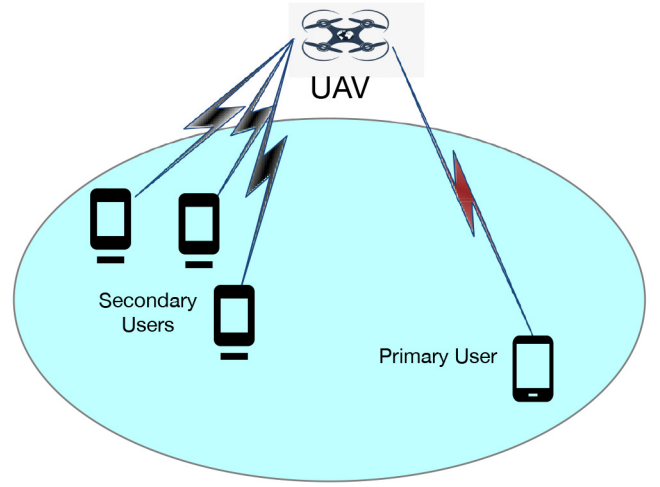


Fig. 1. System model of optimization on EE for UAV-NOMA.

Using NOMA and SIC receiver, the UAV node can serve multiple secondary users simultaneously. Specifically, all data streams are added together with different weights at the transmission node. Accordingly, data streams are decoded layer by layer at each secondary user.

To simplify the following analysis, the continuous trajectory of UAV is divided by equal time slots. The flight altitude of the UAV is fixed as H , and $\mathbf{W} = \{\mathbf{w}[n] = [x[n], y[n]]^T\}$ denotes the trajectory of the UAV node. The total time period of the trajectory is denoted as T , the number of locations is N , and the time interval is $\delta = T/N$.

Let L_p and $L_i = [x_i, y_i]^T$ denote the locations of primary users and the i th secondary user, respectively. Thus, the channel fading power of the destination users is

$$\begin{aligned} g_p[n] &= \rho d_p^{-\alpha}[n] \\ &= \rho \left(H^2 + \|\mathbf{w}[n] - L_p\|^2 \right)^{-\alpha/2} \end{aligned} \quad (1)$$

and

$$\begin{aligned} g_i[n] &= \rho d_i^{-\alpha}[n] \\ &= \rho \left(H^2 + \|\mathbf{w}[n] - L_i\|^2 \right)^{-\alpha/2} \end{aligned} \quad (2)$$

where ρ is the reference power gain with $d_i = 1m$, and $\alpha \geq 2$ is the path-loss exponent.

Note that the line-of-sight channel model without multipath effect is adopted in this article, which results in good agreement with the actual situation for UAV networks, and this assumption is widely used in the existing literature, such as [14] and [35].

Consequently, the downlink NOMA signal transmitted from the UAV node is expressed as

$$x = \sum_{i=1}^K \sqrt{\xi_i} P x_i \quad (3)$$

where P denotes the total signal power, x_i is the original information to the i th secondary user, and ξ_i denotes the power allocation coefficient for x_i . Moreover, $\xi[n]$ is used to represent the power allocation coefficients for the n th time slot,

i.e., $\xi[n] = [\xi_1[n], \xi_2[n], \dots, \xi_K[n]]^T$. and Ξ means the total power allocation factors in all time slots as $\Xi = \{\xi[n], n \in [1, N]\}$. In order to satisfy the power constraint, we obtain

$$\sum_{i \in \Omega} \xi_i[n] = 1. \quad (4)$$

Then, the received wireless signal at the destination users can be given by

$$y_i = \sqrt{g_i}x + n_i = \sqrt{g_i} \sum_{i \in \Omega} \sqrt{\xi_i} P x_i + n_i \quad (5)$$

where $n_i \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise. Without losing generality, we assume that the channel gain increases by index, that is, $g_1 \leq g_2 \leq \dots \leq g_K$. Since the classical NOMA protocol is adopted, the power allocation factors decrease by index, $\xi_1 \geq \xi_2 \geq \dots \geq \xi_K$.

III. PROBLEM FORMULATION

Since the ideal interference cancellation receiver is used in all nodes, the receiver decodes all data streams one by one according to the order of the allocation power factors. In addition, the signal noise ratio (SNR) of the k th data stream is calculated as

$$\begin{aligned} \gamma_{i,k}[n] &= \frac{\xi_k[n]}{\eta_k[n] + \frac{\sigma^2}{g_i[n]P[n]}} \\ &= \frac{\xi_k[n]}{\eta_k[n] + \frac{\sigma^2}{\rho_o P[n]} (H^2 + \|\mathbf{w}[n] - \mathbf{L}_i\|^2)^{\alpha/2}} \end{aligned} \quad (6)$$

where $\eta_k[n]$ represent the interference items at the k th secondary user in the n th time slot, i.e.,

$$\eta_k[n] = \sum_{j=k+1}^K \xi_j[n] \quad \forall n \in [1, N]. \quad (7)$$

From (6), we can see that $\gamma_{i,k}[n]$ is a monotone increasing function. Since $g_i[n]$'s are in an ascending order, we have $\gamma_{i,k}[n] \leq \gamma_{j,k}[n] \quad \forall i \leq j$. According to the SIC receiver, the decoding order is the same as the channel power gains order. Then, the achievable rate of the k th data stream is

$$R_{k,k}[n] = \log_2(1 + \gamma_{k,k}[n]) \quad \forall k \in [1, K], n \in [1, N]. \quad (8)$$

Meanwhile, due to the QoS requirement for each secondary user, we have the following SNR constraint:

$$R_{k,k}[n] \geq C_{th} \quad \forall k \in [1, K], n \in [1, N]. \quad (9)$$

Since $R_{k,k}[n]$ is a monotone increasing function of $\gamma_{k,k}[n]$, (9) is equivalent as

$$\gamma_{k,k}[n] \geq \Gamma \quad \forall k \in [1, K] \quad \forall n \in [1, N] \quad (10)$$

with $\Gamma = 2^{C_{th}} - 1$.

The interference power received by the primary user is

$$I_p[n] = P[n]g_p[n] = \frac{P[n]\rho}{(H^2 + \|\mathbf{w}[n] - \mathbf{L}_p[n]\|^2)^{\alpha/2}}. \quad (11)$$

In each time slot, the sum of achievable transmission rates of all secondary users is

$$R[n] = \sum_{k=1}^K R_{k,k}[n] = \sum_{k=1}^K \log_2(1 + \gamma_{k,k}[n]). \quad (12)$$

Thus, the EE of the n th time slot is

$$EE[n] = \frac{R[n]}{P[n]}. \quad (13)$$

In the following section, our purpose is to maximize the average EE of NOMA-assisted UAV networks with interference constraints. Thus, the optimization problem on EE can be expressed as

$$(P1) : \max_{\mathbf{w}, \xi, P} \frac{1}{N} \sum_{n=1}^N EE[n] \quad (14)$$

$$\text{s.t. } 0 \leq \xi_i[n] \leq 1 \quad (14a)$$

$$\sum_{i \in \Omega} \xi_i[n] = 1 \quad (14b)$$

$$\|\mathbf{w}[n] - \mathbf{w}[n+1]\|^2 \leq (v_m \delta)^2 \quad (14c)$$

$$\gamma_{k,k}[n] \geq \Gamma \quad (14d)$$

$$I_p[n] \leq I_{th} \quad (14e)$$

where v_m denotes the maximum flying speed.

IV. JOINT OPTIMIZATION ALGORITHM

In this section, we will introduce the RL-based solution for (P0). First, with a given UAV trajectory, the Dinkelbach method-based fractional programming is adopted to give the solution of the transmission power as well as the power allocation scheme. Afterward, using the previous power allocation scheme, the successive convex optimization algorithm is adopted in the second stage to update the system parameters. Finally, RL-based optimization is introduced to obtain the best UAV trajectory in terms of mean EE.

A. Subproblem With Given UAV Trajectory

First, with a given UAV trajectory, we focus on the EE in each time slot, i.e., $EE[n]$ as defined in (13). With a definition that

$$R[n] = \sum_{k=1}^K \log_2 \left(1 + \frac{\xi_k[n]}{\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]}} \right) \quad (15)$$

and introducing variables $\eta[n]$, the optimal subproblem can be rewritten as

$$(P1) : \max_{\xi[n], \eta[n], P[n]} EE[n] = \frac{R[n]}{P[n]} \quad (16)$$

$$\text{s.t. } 0 \leq \xi_i[n] \leq 1 \quad (16a)$$

$$\sum_{i \in \Omega} \xi_i[n] = 1 \quad (16b)$$

$$\eta_k[n] = \sum_{j=k+1}^K \xi_j[n] \quad (16c)$$

$$\gamma_{k,k}[n] \geq \Gamma \quad (16d)$$

$$P[n] \leq I_{th}/g_p[n]. \quad (16e)$$

Since the objective function in (16) is a nonlinear fractional optimization problem, it is difficult to be solved. The Dinkelbach algorithm [12] can be adopted to solve the problem with some mathematical transforms.

Proposition 1: The numerator of $EE[n]$, i.e., $R[n]$, is concave w.r.t. $\xi_k[n]$ and $P[n]$, and convex w.r.t. $\eta_k[n]$, respectively.

Proof: Consider a continuous function defined as follows:

$$w = \ln\left(1 + \frac{x}{y}\right), x > 0, y > 0. \quad (17)$$

Applying the first-order partial derivative on w , we obtain

$$\frac{\partial w}{\partial x} = \frac{1}{\left(1 + \frac{x}{y}\right)} \quad (18)$$

and

$$\frac{\partial w}{\partial y} = \frac{-x}{y^2 + xy}. \quad (19)$$

Furthermore, applying the second-order partial derivative on w , yields

$$\frac{\partial^2 w}{\partial x^2} = \frac{-1}{\left(1 + \frac{x}{y}\right)^2} < 0 \quad (20)$$

and

$$\frac{\partial^2 w}{\partial y^2} = \frac{x(x + 2y)}{(y^2 + xy)^2} > 0. \quad (21)$$

Consider another continuous function

$$w = \ln\left(1 + \frac{a}{b + c/z}\right), z > 0. \quad (22)$$

Similarly, w is concave w.r.t. z . According to the definition of $R[n]$, we have

$$\begin{aligned} R[n] &= \sum_{k=1}^K \log_2 \left(1 + \frac{\xi_k[n]}{\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]}} \right) \\ &= \frac{1}{\ln 2} \sum_{k=1}^K \ln \left(1 + \frac{\xi_k[n]}{\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]}} \right). \end{aligned} \quad (23)$$

Similar to (21) and (22), with given $g_i[n]$, it is concluded that $R[n]$ is a concave function w.r.t. $\xi_k[n]$ and $P[n]$ and is convex w.r.t. $\eta_k[n]$.

This is the end of the proof. ■

Applying the first-order partial derivative on $R_{k,k}[n]$ in (8), yields

$$\begin{aligned} &\frac{\partial R_{k,k}[n]}{\partial \eta_k[n]} \\ &= \frac{1}{\ln 2} \frac{-\xi_k[n]}{\left[\left(\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]} \right)^2 + \xi_k[n] \left(\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]} \right) \right]}. \end{aligned} \quad (24)$$

According to Proposition 1, $R_{k,k}[n]$ is a convex function of $\eta_k[n]$. By using the first-order Taylor expansion on $R_{k,k}[n]$, yields

$$\begin{aligned} R_{k,k}[n] &= \frac{1}{\ln 2} \ln \left(1 + \frac{\xi_k[n]}{\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]}} \right) \\ &\geq \frac{1}{\ln 2} \ln \left(1 + \frac{\xi_k[n]}{\eta_k^r[n] + \frac{\sigma^2}{g_k[n]P^r[n]}} \right) + \nabla_k^r (\eta_k[n] - \eta_k^r[n]) \\ &= R_k^L[n] \end{aligned} \quad (25)$$

where $R_k^L[n]$ denotes the lower bound of $R_{k,k}[n]$, and the definition of ∇^r is given as follows:

$$\nabla_k^r = \frac{1}{\ln 2} \frac{-\xi_k^r[n]}{\left[\left(\eta_k^r[n] + \frac{\sigma^2}{g_k[n]P^r[n]} \right)^2 + \xi_k^r[n] \left(\eta_k^r[n] + \frac{\sigma^2}{g_k[n]P^r[n]} \right) \right]} \quad (26)$$

where $\xi_k^r[n]$, $\eta_k^r[n]$, and $P^r[n]$ are the previous r th iteration output results. Note that $R_k^L[n]$ is concave of $\xi_k[n]$ and $P[n]$, respectively, and is linear w.r.t. $\eta_k[n]$, respectively. By substituting (11) into (16d), the maximum value of $P[n]$ is

$$\eta_k[n] + \frac{\sigma^2}{g_k[n]P[n]} \leq \frac{\xi_k[n]}{\Gamma}. \quad (27)$$

Note that (27) is a convex function of $P[n]$. Thus, given r th iteration output results $\xi_k^r[n]$, $\eta_k^r[n]$, and $P^r[n]$, the optimal problem of the $(r + 1)$ th iteration is formulated as

$$\begin{aligned} (P2) : \quad &\max_{\xi[n], \eta[n], P[n]} EE^L[n] = \frac{\sum_{k=1}^K R_k^L[n]}{P[n]} \\ &s.t. \quad (16a), (16b), (16c), (16e), (27). \end{aligned} \quad (28a)$$

Since it is difficult to solve (P1), we present another approximated solution as in (P2). Note that the cost of the transform from (P1) to (P2) is that the first-order Taylor expansion is adopted on the achievable rate as in (25), and the approximated version $R_k^L[n]$ is used instead of the accurate achievable rate $R_{k,k}(n)$. A similar method is widely used in existing works, such as [13] and [36], and simulation results show that the approximation error is negligible.

B. Fractional Programming Using Dinkelbach Method

We can see that all of the constraints in (28) are linear, the numerator is concave w.r.t. $\xi[n]$, $\eta[n]$, and $P[n]$, and the denominator of the objective function is linear. Thus, the Dinkelbach optimization algorithm [12], [13] can be used to solve the problem in (28).

Proposition 2: Let S denoting a compact subset of E^n , where n is the dimension of the Euclidean space. Let $N(x)$ and $D(x)$ are continuous real-value functions, with assumption as follows:

$$D(x) > 0 \quad \forall x \in S. \quad (29)$$

Given the following fractional equation:

$$\max_{x \in S} \frac{N(x)}{D(x)} \quad (30)$$

and

$$F(q) = \max_{x \in S} [N(x) - qD(x)], \text{ for } q \in E^1 \quad (31)$$

we have

- 1) $F(q)$ is continuous and convex over $q \in E^1$.
- 2) $F(q)$ is strictly monotone decreasing w.r.t. q .
- 3) Let

$$q_0 = \frac{N(x_0)}{D(x_0)} = \max_{x \in S} \frac{N(x)}{D(x)} \quad (32)$$

if and only if

$$\begin{aligned} F(q_0) &= F(q_0, x_0) \\ &= \max_{x \in S} [N(x) - q_0 D(x)] = 0. \end{aligned} \quad (33)$$

Proof: The detailed proof is presented in [12].

Furthermore, if $N(x)$ is concave and $D(x)$ is convex, the nonlinear programming in (30) can be solved by the iterative Dinkelbach algorithm as follows.

- 1) Initialization with iteration index $t = 0$, $q_0 = 0$, maximum iteration number T and convergence threshold δ , definition $F(q, x) = N(x) - qD(x)$.
- 2) Given q_t , find the solution of $x_t = \arg \max_x F(q_t, x)$ by using classical convex optimization tools.
- 3) If $[(F(q_t, x_t) > \delta) \text{ and } (t < T)]$ then update parameters with $t = t + 1$ and $q_t = [N(x_t)/D(x_t)]$, goto step 2).
- 4) Output the solution x_t and calculate the programming result with $q(x_t) = [N(x_t)/D(x_t)]$.

With definition that

$$F(q, \xi[n], \eta[n], P[n]) = \sum_{k=1}^L R_k^L[n] - q \cdot P[n] \quad (34)$$

the programming problem in (28) can be rewritten as

$$\begin{aligned} \text{(P3): } & \max_{\xi[n], \eta[n], P[n]} F(q, \xi[n], \eta[n], P[n]) \\ \text{s.t. } & (16a), (16b), (16c), (16e), (27). \end{aligned} \quad (35a)$$

Since $R_k^L[n]$ is a concave function of $\xi_k[n]$ and $P[n]$, respectively, and is linear w.r.t. $\eta_k[n]$, respectively, it can be concluded that the programming problem in (P3) is convex. Moreover, the standard convex optimization tools, such as CVX [37], can be deployed to solve the problem.

By applying the Dinkelbach algorithm on (P2) in (28), we can give the iterative optimization method as described in Algorithm 1.

According to [12], Algorithm 1 is convergent. Followed by the complexity analysis in [38], the computational complexity of Algorithm 1 is

$$C_1 = \mathcal{O}\left\{T_1(N(3K+2))^4(N(K+1))^{1/2} \log(1/\epsilon)\right\} \quad (36)$$

where ϵ denotes the solution accuracy of CVX tools. ■

Algorithm 1 Dinkelbach Method-Based Optimization Algorithm for (P2)

- 1: Initialize the parameters with $t = 0$, $q_0 = 0$, maximum iterative number T_1 and convergence threshold τ_1 .
- 2: **repeat**
- 3: With given q_t , use CVX to solve (P3) and obtain the optimal solution $\xi_{t+1}[n]$, $\eta_{t+1}[n]$, $P_{t+1}[n]$.
- 4: Update iteration index: $t = t + 1$ and update parameters with $q_{t+1} = \frac{R_{t+1}^L[n]}{P_{t+1}[n]}$.
- 5: **until** $t \geq T_1$ or $F(q_{t+1}, \xi_{t+1}[n], \eta_{t+1}[n], P_{t+1}[n]) \leq \tau_1$.
- 6: Output the results q_{t+1} , $\xi_{t+1}[n]$, $\eta_{t+1}[n]$, $P_{t+1}[n]$.

Algorithm 2 Successive Convex Optimization Algorithm for (P1)

- 1: Initialize the parameters with $r = 0$, maximum iterative number T_2 , convergence threshold τ_2 and initial solution $\xi_0[n]$, $\eta_0[n]$, $P_0[n]$, calculate the objective function value $EE_r[n]$ as in (16).
- 2: **repeat**
- 3: With given $\xi_r[n]$, $\eta_r[n]$, $P_r[n]$, use Algorithm 1 to solve problem (P2) and obtain the optimal solution $\xi_{r+1}[n]$, $\eta_{r+1}[n]$, $P_{r+1}[n]$.
- 4: Update iteration index $r = r + 1$ and calculate the increase of the objective function $\chi_r = EE_{r+1}[n] - EE_r[n]$.
- 5: **until** $t \geq T_2$ or $\chi_r \leq \tau_2$.
- 6: Output the solution $\xi_{r+1}[n]$, $\eta_{r+1}[n]$, $P_{r+1}[n]$ and optimal objective function with $EE_{r+1}[n]$.

C. Iterative Optimization Algorithm for (P1)

Since the solution of (P2) can be obtained by Algorithm 1, we can summarize the detailed solution of (P1) by using the successive convex optimization algorithm as Algorithm 2.

Moreover, in each iterative process of Algorithm 2, we can see that $EE_{r+1}[n] \geq EE_r[n] \quad \forall r \quad \forall n$. In another hand, $EE_{r+1}[n]$ is obviously upper bounded. Thus, we conclude that Algorithm 2 is convergent.

The computational complexity of Algorithm 2 can be given by

$$\begin{aligned} C_2 &= T_2 C_1 \\ &= \mathcal{O}\left\{T_2 T_1 (N(3K+2))^4 (N(K+1))^{1/2} \log(1/\epsilon)\right\}. \end{aligned} \quad (37)$$

V. REINFORCEMENT-LEARNING-BASED OPTIMIZATION ON UAV TRAJECTORY

Because the optimization of UAV trajectory is hard to solve directly, we turn to propose a Q -learning-based method to search the action space and select the optimal action according to the reward function. First, the action space is defined as follows: $A = \{0, 1, 2, 3, 4\}$; $\lambda = v_m \delta$ denotes the distance of each movement; the action zero, i.e., $a = 0$, indicates that the UAV keeps the current position; and other actions, such as $a = 1, 2, 3, 4$, denote that the UAV moves to four different directions. Moreover, all users are located within a 2-D space with size $M\lambda \times M\lambda$

$$S = \{s_n = [x[n], y[n]]^T\} \quad (38)$$

with

$$x[n] \in [0, M-1], y[n] \in [0, M-1]. \quad (39)$$

Algorithm 3 Deep RL Optimization Algorithm for (P0)

- 1: Initialize ξ, s , initialize $Q(s, a)$ with arbitrary value, and set iteration index $n = 1$.
- 2: **repeat**
- 3: For each step of iterations
- 4: Employ ϵ -greedy policy to select action a_n , updated state s_{n+1} .
- 5: Given UAV location s_{n+1} , calculate the optimal power allocation factors $\xi[n+1]$ and transmission power $P[n+1]$ as in **Algorithm 2**.
- 6: Observe the reward R_a according to (40).
- 7: Update Q-table as in (41).
- 8: Update iteration index: $n = n + 1$.
- 9: **until** n reaches the maximum iteration number T_3 .

According to the optimization problem in (14), the reward function of each time slot R_a is

$$R_a = EE[n] \quad (40)$$

where the definition of $EE[n]$ is given in (16)

$$Q_{n+1}(s_n, a_n) = (1 - \theta)Q_n(s_n, a_n) + \theta \left[R_n + \beta \max_{a \in A} Q_n(s_{n+1}, a) \right] \quad (41)$$

where $\theta \in (0, 1]$ is the learning rate, and $\beta \in [0, 1]$ is the discount factor. In each step, the UAV action is updated according to the ϵ -greedy policy, i.e., the optimal action is selected with probability ϵ , while the other actions are selected equally with total probability $(1 - \epsilon)$. The detailed solution procedure for (P0) is given in Algorithm 3.

The computational complexity of Algorithm 3 is calculated by

$$C_3 = 5M^2C_2 = \mathcal{O}\left\{5M^2T_2T_1(N(3K+2))^4(N(K+1))^{1/2}\log(1/\epsilon)\right\}. \quad (42)$$

Furthermore, in each iterative process of Algorithm 3, we can see that $Q_{n+1}(s_n, a_n) \geq Q_n(s_n, a_n) \forall n$. On the other hand, $Q_{n+1}(s_n, a_n)$ is obviously upper bounded. Thus, we conclude that Algorithm 3 is convergent.

VI. SIMULATION RESULTS

The proposed optimization method is verified by numerical results.

Without special comments, some parameters in the reference configuration are listed as follows: the target area is a square with 100×100 m; the number of secondary users is $K = 3$; the locations of the secondary users are $[20, 20]$ m, $[40, 40]$ m, and $[60, 60]$ m; the initial location is $w_0 = [80, 0]$ m; the location of the primary user is $[0, 50]$ m; the reference power gain is $\rho = 10^{-3}$; the UAV trajectory altitude is $H = 20$ m; the path-loss exponent is $\alpha = 2$; the iteration numbers are set as $T_1 = 8$ and $T_2 = 3$; and the convergence thresholds are $\tau_1 = 10^{-3}$ and $\tau_2 = 10^{-1}$, the threshold of the epsilon-greedy algorithm is $\epsilon = 0.9$; the discount factor is $\theta = 0.95$; the learning rate is $\beta = 0.4$; and the total number of time slots is set as $N = 20000$.

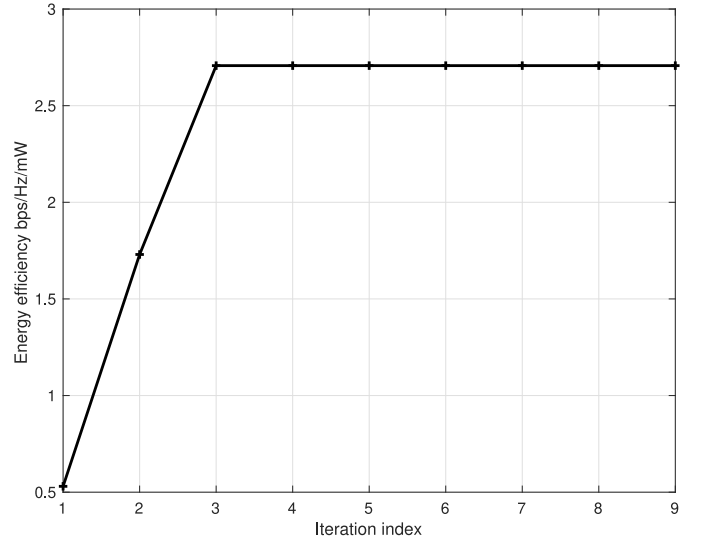


Fig. 2. Convergence performance of Algorithm 1 for (P2).

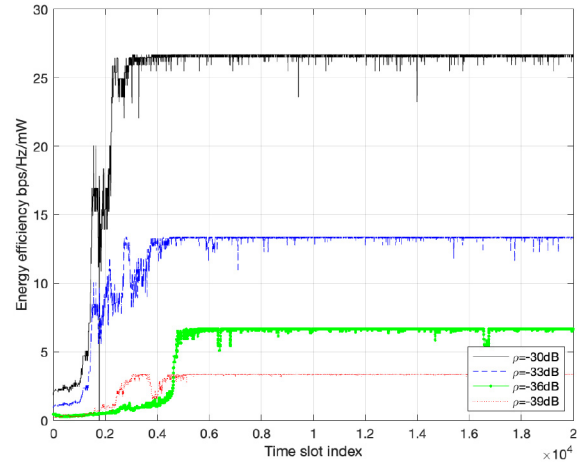


Fig. 3. Convergence performance of Algorithm 3 with different ρ .

The convergence of Algorithm 1 for (P2) is depicted in Fig. 2. We can see from this figure that Algorithm 1 converges rapidly w.r.t. the iteration index. When the iteration number is larger than 3, the EE becomes stationary, which can verify the convergence of the Dinkelbach method-based iteration optimization algorithm in Algorithm 1. Also, the convergence performance of the RL-based optimization algorithm given in Algorithm 3 is presented in Fig. 3, where the reference fading gain ρ changes from -30 to -39 dB. We can see from this figure that the EE increases rapidly for the first 5000 time slots and then remains stationary, which verifies the convergence performance of the proposed method.

The impacts of UAV altitude H on mean EE are presented in Fig. 4. The UAV altitude H changes from 20 to 50 m. We can see that with $H = 20$ m, when the time slot number is large enough, the mean EE reaches about 23 bps/Hz/mW. While with $H = 50$ m, the mean EE drops quickly down to 7 bps/Hz/mW. A fixed trajectory scheme, which surrounds the boundary of the target area and uses the same power allocation algorithm as given in Algorithm 1, is adopted as a performance benchmark. The proposed algorithm shows

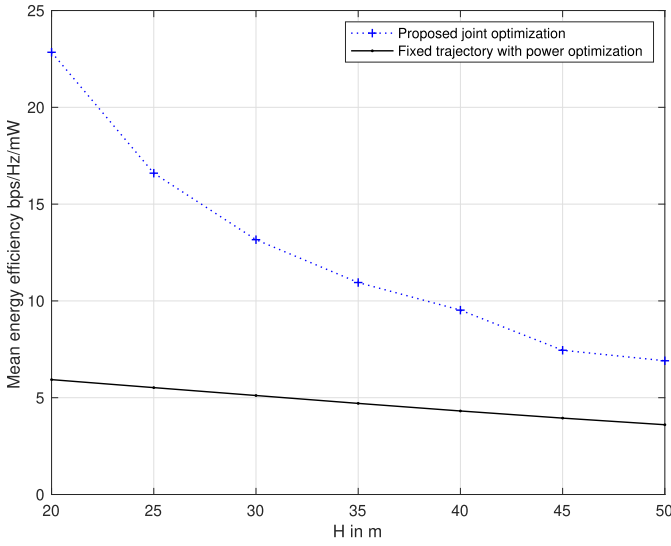
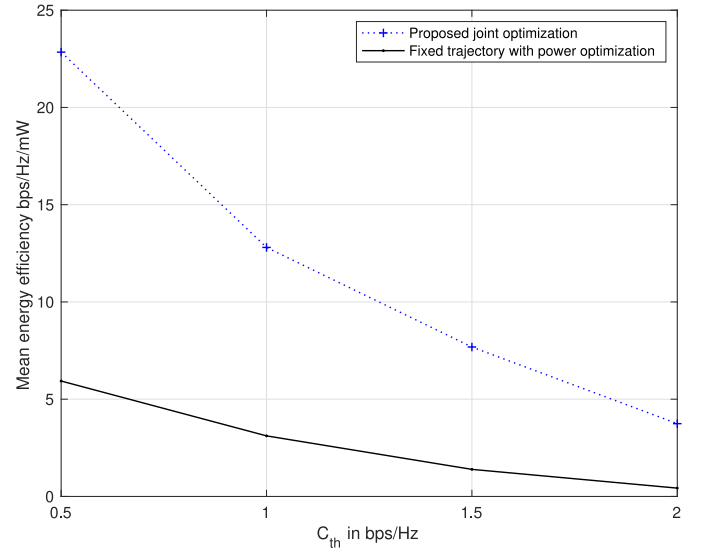
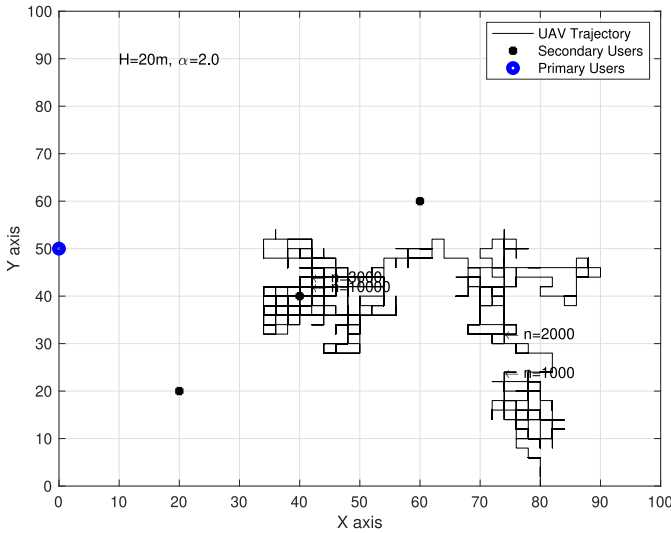
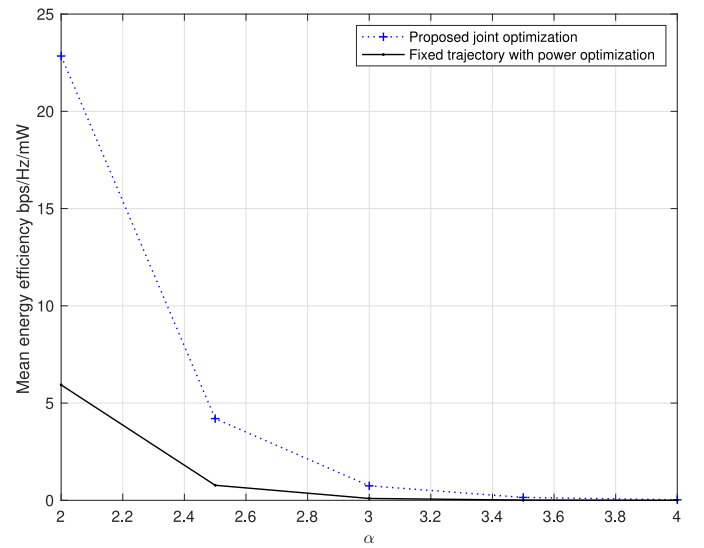


Fig. 4. Performance comparison on mean EE.

Fig. 6. Effects of capacity threshold C_{th} on mean EE.Fig. 5. UAV trajectory with $H = 20$ m.Fig. 7. Effects of path fading exponent α on mean EE.

remarkable performance gain compared with the baseline in all setup cases. Furthermore, the UAV trajectory with $H = 20$ m is given in Fig. 5. One can see that by using of about 3000 explores, the UAV trajectory becomes stationary within an area of about $10 \text{ m} \times 10 \text{ m}$.

Fig. 6 shows the effects of capacity threshold C_{th} on mean EE, where the capacity threshold C_{th} changes from 0.5 to 2.0 bps/Hz. It can be concluded that the larger the capacity threshold C_{th} is, the smaller the stationary EE is. Specifically, with $C_{th} = 0.5$ bps/Hz, the mean EE reaches 23 bps/Hz/mW when the slot number is 20000. While the corresponding result is only 4 bps/Hz/mW with $C_{th} = 2.0$ bps/Hz, i.e., 82.6% reduction in terms of mean EE. The reason is that when the capacity threshold C_{th} becomes larger, more transmission power is used to support the minimum capacity of each secondary user. Afterward, the feasible solution is compressed and the value of the objective function is decreased.

The effects of path fading exponent α on mean EE are given in Fig. 7, where the path fading exponent α changes from

2 to 4. We can observe that when α changes larger, the EE reduces from 23 to smaller than 1 bps/Hz/mW. The reason is that when the path fading exponent becomes larger, more energy is consumed to achieve the same transmission rate. Thus, the mean EE becomes smaller rapidly. Also, a significant performance gain can be obtained by the proposed iterative optimization algorithm compared with the fixed trajectory scheme.

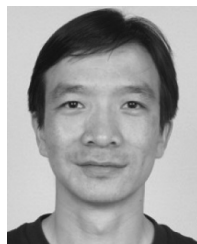
VII. CONCLUSION

The combination of NOMA and UAV can be adopted in industrial IoT to improve the wireless connections and spectrum efficiency. On the condition of interference constraint and the minimum rate of each secondary user, we propose an iterative RL-based algorithm on joint optimization of UAV trajectory and power allocation scheme for NOMA protocol. First, with a given UAV trajectory, the Dinkelbach method-based fractional programming is adopted to obtain the optimal transmission power and the power allocation scheme. With the

help of the previous power allocation scheme, the successive convex optimization algorithm⁷ is used in the second stage to update the system parameters. Finally, RL-based optimization is introduced to obtain the best UAV trajectory in terms of mean EE.

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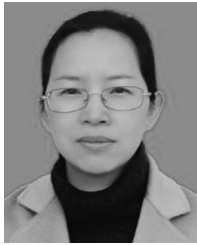


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